

Amendment No. 4 (04/2026)

To

AIS 040 Rev 1: 2015: Electric Power Train Vehicles - Method of Measuring the Range

1.0 Scope

Add following note below the scope of the standard

Note :With effective date of applicability of AIS-175 for ICE vehicles, Range for Electric Power Train Vehicles of category M1 & M2 with GVW up to 3.5 T shall be measured as per the procedure laid down in AIS-175 and for categories L, M2 with GVW above 3.5 T, M3 & N shall be as per AIS-040 (Rev.1)

2.0 Clause No 5.1.1

Substitute following text for existing text:

5.1.1 The driving cycle shall be the World-harmonised Motorcycle Test Cycle (WMTC) Stage 2 for L1 and L2 category vehicles as per AIS 137 Part 1 and details as per below table. The cycles RST25 and RST45 shall be as per Amendment 2 to AIS 049 (Rev.1).

For vehicles of category L5, L7 driving cycle shall be Indian Driving Cycle (IDC) as given in Annexure-II of CMVR.

Sr. No	Category of Vehicle	Max Speed	Test Cycle
1	L1 and L2	$V_{max} \leq 25 \text{ km/h}$	Part 1, RST25, followed by Part 1, RST25
		$25 \text{ km/h} < V_{max} \leq 50 \text{ km/h}$	Part 1 reduced speed followed by Part 1 reduced speed, if maximum design speed is $\leq 50 \text{ km/h}$ Part 1, RST45, followed by Part 1, RST45, if maximum design speed is $\leq 45 \text{ km/h}$
		$50 \text{ km/h} < V_{max} < 100 \text{ km/h}$	Part 1 reduced speed followed by Part 1 reduced speed
		$100 \text{ km/h} \leq V_{max} < 115 \text{ km/h}$	Part 1 reduced speed followed by Part 2 reduced speed

		$115 \text{ km/h} \leq V_{\max} < 130 \text{ km/h}$	Part 1 followed by Part 2
		$130 \text{ km/h} \leq V_{\max} < 140 \text{ km/h}$	Part 1 followed by Part 2 followed by Part 3 reduced speed
		$V_{\max} \geq 140 \text{ km/h}$	Part 1 followed by Part 2 followed by Part 3

3.0 Clause No 5.3.1

Add following note at the end

Note: If vehicle has multiple driving modes and vehicle is not able to follow the specified test cycle speed as mentioned in Clause No 5.1.1 in default mode then the test shall be carried out in the closest mode wherein vehicle follows the specified WMTC test cycle speed.

4.0 Add following paragraph at the end of paragraph 5.5

For M1 category vehicles additional range measurement shall be carried out with Air-conditioning system in operation as per the procedure given in AIS 213:2024

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

9th April 2026

Amendment No. 3 (04/2024)
To
AIS 040 (Rev 1): 2015: Electric Power Train Vehicles - Method of Measuring
the Range

1.0 Clause No 5.1.2

Substitute following text for existing text:

5.1.2 The driving cycle shall be Modified Indian Driving Cycles (MIDC) each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) as given in Annexure-IV B of CMVR for vehicles of category M1, M2 (with GVW up to 3500 kg).

The driving cycle shall be, Part-I of the Modified Indian Driving Cycle as given in Annexure-IV B of CMVR, for vehicles of category N.

The driving cycle shall be Delhi Driving Cycle as given in AIS-049 (Rev 1):2016, as amended and revised from time to time, for M2 (with GVW above 3500 kg) and M3 category of vehicles.

2.0 Clause No 5.5

Delete last paragraph

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UNDER
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SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

4th April 2024

Amendment 2 (11/2021)
To
AIS 040 Rev 1:2015 : Electric Power Train Vehicles - Method of Measuring the Range

1.0 Page No 1/6 Clause No 5.1.2

Substitute following text for existing text:

5.1.2 For vehicles of category M1, M2 (with GVW up to 3500 kg), two separate range measurement tests shall be carried out as per driving cycles given below.

For vehicles of category M1, M2 (with GVW up to 3500 kg)	The driving cycle shall be a) Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR, for measurement of range in km. b) Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) of the modified Indian driving cycle (MIDC) as given in Annexure-IV B of CMVR, for measurement of “Urban + Extra-Urban- range” in km
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The driving cycle shall be, Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR, for vehicles of category N.

The driving cycle shall be Delhi Driving Cycle as given in AIS-049:2003, as amended and revised from time to time for M2 (with GVW above 3500 kg) and M3 category of vehicles.

2.0 Page No 2/6 Clause No 5.3.1

Substitute following text with existing text:

5.3.1 Principle

The test method described hereafter permits to measure the range of the Electric Power Train Vehicles expressed in km.

If the Electric Power Train Vehicle has multiple driving modes,

where default mode is –

- a) Available, test shall be carried out in default mode.
- b) Not available, then test shall be carried out in worst case mode agreed between vehicle manufacture and test agency.

3.0 Page No 4/6 Clause No 5.5 paragraph 2

Add below text above paragraph 2

For L1 category of vehicles, automatic head lamp and for L2 category of vehicles, automatic head lamp or daytime running lamps shall be ON, during the test cycle.

For vehicles of other than L1 and L2 category, if DRL is installed on vehicle, DRL may be switched ON during test cycle as per vehicle manufacture's recommendation.

The vehicle tested shall be equipped with the daytime running lamp system that has the highest electrical energy consumption among the daytime running lamp systems, which are fitted by the manufacturer to vehicles.

4.0 Page No 4/6 Clause No 5.5 paragraph 2

Add below text at the end of last paragraph.

For vehicles of category M1, M2 (with GVW up to 3500 kg), measured D of the covered distance in km with Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR, is the electric range, adopted as the type approved value of the electric vehicle.

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AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

18th November 2021

AMENDMENT 1 19th December 2017

To

AIS-040 (Rev.1): Electric Power Train Vehicles - Method of Measuring the Range

1.0 Page No. 4, Clause 5.5, paragraph 2

Substitute following text for the existing:

The end of test criteria shall be when the vehicle is not able to meet the target curve up to 50 km/h, (or 85% of the maximum speed of the driving cycle or 85% of the maximum speed of the vehicle for L category of vehicles only) or when an indicator from the standard on-board instrumentation is given to the driver to stop the vehicle.

Then the vehicle shall be slowed down to 5 km/h by deactivating the accelerator control, without touching the brake control and then stopped by braking.

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UNDER
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SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
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19th December 2017

AUTOMOTIVE INDUSTRY STANDARDS

**Electric Power Train Vehicles -
Method of Measuring the Range**

(Revision 1)

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

February 2015

Status chart of the Standard to be used by the Purchaser for updating the record.

Sr. No.	Corrigenda	Amend- ment	Revision	Date	Remark	Misc.

General Remarks :

INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of Standards and development of test facilities in parallel when the work of preparation of Standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the Standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standard Committee (AISC) vide order no. RT-11028/11/97-MVL dated September 15, 1997. The Standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CTSC) after approval, The Automotive Research Association of India, (ARAI), Pune, being the secretariat of the AIS Committee, has published this Standard. For better dissemination of this information, ARAI may publish this standard on their website.

This standard prescribes the requirements for the measurement of range of battery operated vehicle.

Considerable assistance has been taken from UN R101 Rev 3(Supplement 1 to the 01 series of amendments - Date of entry into force: 27 January 2013): Uniform Provisions concerning the approval of Passenger Cars powered by an Internal Combustion Engine only, or powered by a Hybrid Electric Power Train with regard to the measurement of the Emission of Carbon Dioxide and Fuel Consumption and/or the measurement of Electric Energy Consumption and Electric Range, and of categories M1 and N1 vehicles powered by an Electric Power Train only with regard to the measurement of Electric Energy Consumption and Electric Range.

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annex A and Annex B respectively.

Electric Power Train Vehicles - Method of Measuring the Range

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Electric Power Train Vehicles - Method of Measuring the Range

1.0 SCOPE

This standard specifies the method for measurement of range expressed in km for L, M and N categories of Electric Power Train Vehicles as defined in Rule 2 (u) of CMVR.

2.0 DEFINITIONS

Refer AIS-049:2003 as amended and revised from time to time, for the definitions.

3.0 VEHICLE PREPARATION

Vehicle preparation shall be as per details given in para 3 of AIS-039 (Rev. 1).

4.0 CLIMATE CONDITION

Climate conditions shall be as per details given in para 3.4 of AIS-039 (Rev.1).

5.0 TEST PROCEDURE

5.1 Test Sequence

5.1.1 The driving cycle shall be the Indian Driving Cycle (IDC) as given in Annexure-II of CMVR for all L category vehicles.

5.1.2 The driving cycle shall be Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR for M1, M2 (with GVW upto 3500 kg) and N category of vehicles.

The driving cycle shall be Delhi Driving Cycle as given in AIS-049:2003 as amended and revised from time to time (Rev 1) for M2 (with GVW above 3500 kg) and M3 category of vehicles.

5.1.3 In cases where the vehicle does not reach the required acceleration during driving, the accelerator/speed control shall remain fully activated until the reference curve has been reached again

5.2 Power setting of the chassis dynamometer

The procedure prescribed in the document MoRTH/TAP/CMVR-115/116 as amended from time to time shall be adopted. Reference mass shall be taken as defined in AIS-049:2003 as amended and revised from time to time.

5.3 Test Method

5.3.1 Principle

The test method described hereafter permits to measure the range of the Electric Power Train Vehicles expressed in km.

5.3.2

Parameters, Units and Accuracy of Measurements Parameter	Unit	Accuracy	Resolution
Time	s	+/- 0.1 s	0.1 s
Distance	m	+/-0.1 per cent	1m
Temperature degrees	C	+/- 1 degree C	1 degree C
Speed	km/h	+/- 1 per cent	0.2 km/h
Mass	kg	+/-0.5 per cent	1 kg
Electricity balance	Ah	+/-0.5 per cent	0.3 per cent
Where accuracy is specified in %, it is the % of the measured value			

5.4 Operation Modes

The test method includes the following steps:

- a) Initial charge of the Rechargeable Energy Storage System (REESS)
- b) Application of the cycle and measurement of the range.

Between the steps, if the vehicle has to be moved, it is pushed to the following test area (without regenerative recharging).

5.4.1 Initial charge of the Rechargeable Energy Storage System (REESS)

Charging the REESS consists of the following procedures:

Note: "Initial charge of the REESS" applies to the first charge of the REESS, at the reception of the vehicle. In case of several combined tests or measurements, carried out consecutively, the First charge carried out shall be an "initial charge of the REESS" and the following may be done in accordance with the "normal overnight charge" procedure.

5.4.1.1 Discharge of the Rechargeable Energy Storage System (REESS)

5.4.1.1.1 The procedure starts with the discharge of the REESS of the vehicle while driving (on the test track, on a chassis dynamometer, etc.) at a steady speed of 70 per cent +/-5 percent from the maximum thirty minutes speed of the vehicle.

5.4.1.1.2 Stopping the discharge occurs:

- (a) When the vehicle is not able to run at 65 per cent of the maximum thirty minutes speed;
- (b) Or when an indication to stop the vehicle is given to the driver by the standard onboard instrumentation; or
- (c) After covering the distance of 100 km.

5.4.1.2 Application of a normal overnight charge

The Rechargeable Energy Storage System (REESS) shall be charged according to the following procedure.

5.4.1.2.1 Normal overnight charge procedure

The charging is carried out:

- (a) With the on board charger if fitted; or
- (b) With an external charger recommended by the manufacturer using the charging pattern prescribed for normal charging;
- (c) In an ambient temperature comprised between 20 °C and 30 °C.

Note: If the ambient condition cannot be met at the time of the test, then based on mutual agreement between test agency and vehicle manufacturer, requirement of ambient temperature condition can be waived.

This procedure excludes all types of special charges that could be automatically or manually initiated like, for instance, the equalisation charges or the servicing charges. The manufacturer shall declare that during the test, a special charge procedure has not occurred.

5.4.1.2.2 End of charge criteria

The end of charge criteria corresponds to a charging time not exceeding twelve hours, except if a clear indication is given to the driver by the standard instrumentation that the REESS is not yet fully charged.

In this case,

3 X claimed battery capacity (Wh)
the maximum time is =-----
mains power supply (W)

5.5 Application of the Cycle and Measurement of the Range

The test sequence shall be followed as per 5.1 above.

The end of test criteria shall be when the vehicle is not able to meet the target curve up to 50 km/h , (or 85% of the maximum speed of the driving cycle or 85% of the maximum speed of the vehicle for L1 category of vehicles only) or when an indicator from the standard on-board instrumentation is given to the driver to stop the vehicle. Then the vehicle shall be slowed down to 5 km/h by deactivating the accelerator control , without touching the brake control and then stopped by braking.

When the vehicle does not reach the required acceleration or speed of the test cycle, the accelerator control shall remain fully activated until the reference curve has been reached again.

To respect human needs, up to three interruptions shall be permitted between test sequences of not more than 15 minutes in total.

At the end, measure D of the covered distance in km is the electric range of the electric vehicle. It shall be rounded to nearest whole number.

6.0 TECHNICAL SPECIFICATIONS

The details of technical specification, approvals of changes in specification shall be as per para 5.0 of AIS-049:2003 as amended and revised from time to time.

7.0 TRANSITIONAL PROVISIONS

- 7.1 General guidelines for transitional provisions for this standard shall be as per AIS-000, as amended from time to time, as applicable with the following requirement.
- 7.2 For L category vehicles type approvals issued for compliance to AIS-040-2003 shall be extended for approval to of AIS-040 (Rev1)-2015. A retest may be performed at the request of manufacturer.
- 7.3 For M and N category vehicles re-testing shall be carried out for compliance to AIS 040 (Rev 1): 2015 from the date of notification. Manufacturers may request testing as per this standard from the date of adoption of this standard in CMVR-TSC.

(See Introduction)

**COMPOSITION OF AISC PANEL ON
ELECTRIC POWER TRAIN VEHICLES- CONSTRUCTION
AND FUNCTIONAL SAFETY REQUIREMENTS***

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* At the time of approval of this Automotive Industry Standard (AIS)

ANNEX B
(See Introduction)
COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

Chairperson	
Mrs. Rashmi Urdhwareshe	Director The Automotive Research Association of India, Pune
Members	Representing
Representative from	Ministry of Road Transport and Highways (Dept. of Road Transport and Highways), New Delhi
Representative from	Ministry of Heavy Industries and Public Enterprises (Department of Heavy Industry), New Delhi
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Shri Shrikant R. Marathe	Former Chairman, AISC
Shri N. K. Sharma	Bureau of Indian Standards, New Delhi
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Shri Uday Harite	Automotive Components Manufacturers Association of India, New Delhi

Member Secretary
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General Manager
The Automotive Research Association of India, Pune

* At the time of approval of this Automotive Industry Standard (AIS)